C16 Organic Analysis – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Mass Spectrometry and IR **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Decimal places given by high resolution MS				4 or more
2	Use of high resolution MS				Distinguish compounds with the same integer M_r
3	Molecular ion peak				M^+, highest *m/z*
4	What absorbs infrared radiation				Vibrating bonds
5	x-axis of an IR spectrum				Wavenumber (cm^{-1})
6	y-axis of an IR spectrum				Transmittance (%)
7	Fingerprint region				Below 1500 cm^{-1}
8	Use of the fingerprint region				Identify a compound by comparison
9	Three greenhouse gases				CO_2, CH_4, H_2O
10	Radiation absorbed by greenhouse gases				Infrared

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C16 Organic Analysis – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: IR Absorptions **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	O–H (alcohol)				3230–3550 cm^{-1}, broad
2	O–H (carboxylic acid)				2500–3000 cm^{-1}, very broad
3	N–H (amine)				3300–3500 cm^{-1}
4	C–H				2850–3300 cm^{-1}
5	C≡N				2220–2260 cm^{-1}
6	C=O				1680–1750 cm^{-1}
7	C=C				1620–1680 cm^{-1}
8	C–O				1000–1300 cm^{-1}
9	Absorptions for an aldehyde or ketone				C=O only (no O–H)
10	Absorptions for a carboxylic acid				Very broad O–H and C=O

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.